

1. Administrative & Study Identification

Full Study Title: A Phase IIIb, Multicenter, Single-arm Study Assessing the Effectiveness, Safety and Patient Reported Outcomes of a 36-week Refill Exchange Regimen for the Port Delivery System With Ranibizumab in Patients With Neovascular Age-related Macular Degeneration **Study Title:** A Study of 36-Week Refill Exchanges of Port Delivery System (PDS) With Ranibizumab in nAMD **Short Title/ Acronym:** Sightspire **Protocol Number:** MR45625 / 2024-516924-32-00 **NCT Number:** NCT06847542 **Posting ID:** 608329697416890831 **Trial Status:** Recruiting **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Port Delivery System (PDS) with ranibizumab 100 mg/mL (Trade names: Lucentis, Susvimo, Cimerli, Byooviz) **Phase of Development:** Phase IIIb (PHASE3) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effectiveness, safety, and Patient Reported Outcomes (PROs) of the Port Delivery System (PDS) with ranibizumab 100 mg/mL refilled every 36 weeks (Q36W) in participants with neovascular age-related macular degeneration (nAMD). **Secondary Objectives:** To assess long-term anatomical outcomes (via OCT), sustained Best Corrected Visual Acuity (BCVA), and the incidence of ocular and non-ocular adverse events associated with the extended 36-week refill regimen.

2.2 Background & Rationale Neovascular age-related macular degeneration (nAMD), or wet AMD, is characterized by the growth of abnormal, leaky blood vessels in the central back part of the eye, driven by vascular endothelial growth factor (VEGF). While regular intravitreal (IVT) injections of anti-VEGF therapies like ranibizumab are standard, the high treatment burden often limits real-world adherence and visual outcomes. The Port Delivery System (PDS) is an eye implant designed to continuously release ranibizumab. This study investigates an extended 36-week (Q36W) refill exchange regimen to determine if it can safely maintain efficacy while significantly reducing the treatment burden and improving patient convenience.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **\$N\$:** 250 **Number of Sites:** Multicenter (Global, including sites in Switzerland, Greece, Italy, France, Poland, and South Korea) **Estimated Study Start:** Late 2025 **Estimated Primary Completion:** 12-Oct-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Initial diagnosis of nAMD within 24 months prior to screening. 2. Previous treatment with at least 3 anti-VEGF IVT injections for nAMD per standard of care within 6 months prior to screening. 3. Demonstrated response to prior anti-VEGF IVT treatment since diagnosis. 4. Availability of historical Visual Acuity (VA) and OCT image data obtained at or after nAMD diagnosis and prior to the first anti-VEGF treatment. 5. BCVA of 34 letters or better using ETDRS chart at a starting distance of 4 meters at screening and enrollment visits.

4.2 Exclusion Criteria 1. *Either Eye:* Any active or history of uveitis, keratitis, scleritis, endophthalmitis, or chronic blepharitis; active thyroid eye disease. Prior treatment with brolucizumab or gene therapy for nAMD. CNV due to other causes or masquerading lesions. 2. *Study Eye:* History of vitrectomy surgery, submacular surgery, or other surgical intervention for AMD; previous treatment with corticosteroid intravitreal injection; previous intraocular device implantation. History of vitreous hemorrhage or rhegmatogenous retinal detachment. Subretinal hemorrhage involving the fovea center or subfoveal fibrosis/atrophy. 3. *Systemic:* Pregnant or breastfeeding women, concurrent participation in another therapeutic clinical trial for wet AMD. History of stroke or MI within the last 3 months. Active cancer within 12 months of enrollment (except for appropriately treated in situ cervix, non-melanoma skin, and prostate cancer).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ranibizumab 100 mg/mL refilled every 36 weeks (Q36W). **Route of Administration:** Surgical implantation of the Port Delivery System (PDS) into the eye, followed by in-clinic intravitreal refill exchanges. **Duration of Treatment:** The implant is intended to remain life-long in the eye unless removed for medical reasons; study evaluation spans approximately 3 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard ocular medications/drops unassociated with VEGF inhibition or intraocular steroids. **Prohibited:** Concurrent systemic or intravitreal anti-VEGF therapies in the study eye (unless administered as protocol-defined rescue therapy), intraocular corticosteroids, and investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical/Ocular History, OCT, Visual Acuity (VA) Baseline
Baseline	Day 0	Surgical Implantation of PDS device, Initial Ranibizumab Fill
Interim	Q36W (Every 36 Weeks)	PDS Refill Exchange, Safety Ocular Exam, VA Assessment, OCT, PRO Collection
End of Study	2028	Final Visual Acuity, Exit Ocular Exam, Final PROs, Device Integrity Check

(Note: Interim safety monitoring visits may be scheduled between refill exchanges as per protocol.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change From Baseline in Best Corrected Visual Acuity (BCVA) Score Averaged Over Weeks 68 and 72, as Assessed Using the Early Treatment Diabetic Retinopathy Study (ETDRS) Chart at a Starting Distance of 4 Meters (Range 0 to 100 letters). Higher scores indicate improvement in visual acuity (VA). **Measurement Method:** Standardized Best Corrected Visual Acuity (BCVA) charts (ETDRS), standardized PRO questionnaires, and ophthalmic examinations.

7.2 Secondary & Exploratory Endpoints

- * Change From Baseline in BCVA Score Over Time.
- * Percentage of Participants Who Lose ≤ 15 , ≤ 10 , ≤ 5 , or Gain ≥ 0 Letters in BCVA Score From Baseline Over Time.
- * Change From Baseline in Center Point Thickness (CPT).
- * Incidence, severity, and causality of ocular adverse events (e.g., endophthalmitis, vitreous hemorrhage, implant dislocation).
- * Number of supplemental (rescue) ranibizumab injections required before the 36-week refill mark.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Specialized ophthalmic monitoring for device-related complications (e.g., conjunctival erosion, retinal detachment) and drug-related AEs via slit-lamp and dilated fundus examinations. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for all sight-threatening events (e.g., endophthalmitis). **Data Monitoring Committee (DMC):** External independent data monitoring committee to periodically review device safety and ocular event rates.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary effectiveness and PRO analyses; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Powered to assess whether

a 36-week interval provides non-inferior visual outcomes and acceptable safety compared to historical shorter-interval controls. **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) for continuous efficacy endpoints (BCVA, CPT) and Multiple Imputation as necessary.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Centralized reading centers for OCT and VA data to ensure unbiased evaluation; standard onsite clinical monitoring for surgical and refill procedures. **Audit Trail:** eCRF locking, certified surgical training logs, and specific device accountability logs.

11. Study Identifiers & Governance

Primary ID: MR45625 **Registry Number:** NCT06847542 **Posting ID:**

608329697416890831 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** A Study of 36-Week Refill Exchanges of Port Delivery System (PDS) With Ranibizumab in nAMD **Official**

Regulatory Title: A Phase IIIB, Multicenter, Single-arm Study Assessing the Effectiveness, Safety and Patient Reported Outcomes of a 36-week Refill Exchange Regimen for the Port Delivery System With Ranibizumab in Patients With Neovascular Age-related Macular Degeneration

12. Clinical Framework (nAMD Specific)

Primary Condition: Neovascular Age-related Macular Degeneration (nAMD) **Diagnosis Threshold:** Initial diagnosis within 24 months

prior to screening with proven response to prior anti-VEGF therapy **Medical Methodology:** Continuous Intraocular Release via Port Delivery System (PDS) **Optimization Target:** 36-Week (Q36W) Refill Exchange Regimen

13. Study Procedures & Methodology

Management Pathway: Surgical implantation followed by structured, outpatient refill exchange procedures. **Education Protocol (HCP):** Specialized surgical training for PDS implantation, device handling, and precision refill-exchange technique. **Education Protocol (Patient):** Post-operative surgical care, symptom awareness for ocular emergencies (e.g., sudden vision loss, pain, redness), and self-reporting guidance. **Quality Audit Mechanism:** Centralized imaging reading centers (OCT) and strict visual acuity protocol adherence tracking.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up up to study completion in October 2028
Onsite Visit Cadence: Baseline surgery, mandatory Q36W refill exchange visits, and protocol-defined interim ocular safety checks. **Remote Monitoring Frequency:** Periodic follow-up for PRO completion and general well-being. **Checklist Review Interval:** Review of patient-reported visual changes at every clinical interaction.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				